



Natural history of generalized motor seizures: A retrospective analysis[☆]

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ABSTRACT

Purpose: This study aims to characterize the natural history of generalized motor seizures through longitudinal stratification of patient-reported clinical seizures into high, medium and low rates of generalized motor seizures (also known as generalized tonic-clonic seizures or GTCs).

Methods: From 2007 to 2018, 1.4 million seizures were recorded by 12,402 SeizureTracker.com users that met inclusion/exclusion criteria. The number of GTCs per year since the first seizure diary entry was calculated for each user and categorized as: Low (0 GTCs/year), Medium (1–2 GTCs/year), or High (≥ 3 GTCs/year) GTC rates. **Results:** Kaplan–Meier survival curves for the time until exiting the initial category were computed. There was a global difference between risk groups ($p < 0.01$). Further pairwise log rank tests revealed a difference between each pair of risk groups ($p < 0.01$). At 3 years, 40.8% of people initially presenting with high GTC rates remained in their initial category, while 77.3% of people initially presenting with low GTC rates remained in their initial category.

Conclusion: A patient with a low rate of GTCs is likely to remain at low risk for future GTCs, whereas higher GTC rate patients (at least one GTC/year) may leave their initial risk stratification. Thus, yearly re-assessment may be prudent when considering risk of further GTCs. Given the association between higher yearly rates of GTCs with increased SUDEP risk and morbidity in epilepsy, further validation of these findings is important for prognostication.

1. Introduction

SeizureTracker.com is one of the world's largest online and mobile patient-reported seizure diary databases [1]. It affords a unique view into self-reported seizures from a large and diverse group of patients [2]. For instance, patients self-report different seizure types, therefore it may be possible to learn what kinds of patterns are seen across time with specific seizure types of interest and seizure frequency.

Active seizure frequency has been associated with greatest burden of epilepsy-related disabilities, including social stigma, mental health comorbidities, poor quality of life, economic burden, and increased mortality [3–6]. Sudden Unexpected Death in Epilepsy (SUDEP) risk for adults is 1.2/1000 patient-years [7], and can be up to 18 times greater in patients with frequent generalized seizures. A recent meta-analysis identified the following risk factors: presence of generalized tonic-clonic (GTC) seizures, frequency of GTCs, and absence of seizure freedom [7]. In another meta-analysis, the greatest risk factor was

presence of three or more GTCs per year [8]. These risk factors have not been systematically studied on a population level to evaluate how the natural history of how an individual's risk evolves over time.

This study aims to stratify SeizureTracker.com patient diaries into High, Medium and Low yearly rates of GTCs based on self-report. Furthermore, we evaluate how risk for SUDEP changes over time depending on initial yearly rates.

2. Material and methods

2.1. Study population

Deidentified and unlinked data were extracted on March 12, 2018, from SeizureTracker.com in accordance with Beth Israel Deaconess Medical Center Institutional Review Board (IRB). At the time of export, users recorded 1.6 million seizures and clinical data between December 1, 2007 and February 28, 2018. Of the documented seizures, 1.4 million

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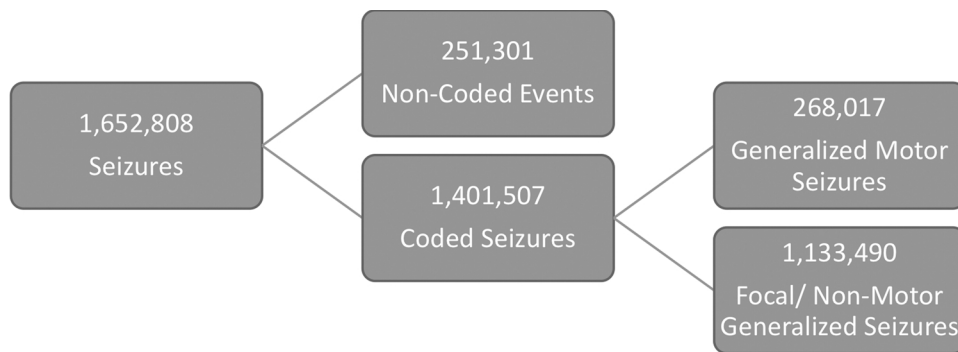


Fig. 1. Schema illustrating SeizureTracker.com data pre-processing. From December 2007 to March 2018, 1.6 million seizures were recorded on SeizureTracker.com. Of these, 251,301 seizures were excluded as they were not properly recorded with a recognizable seizure description: “other”, “unknown”, or missing. Of the described seizures, 268,017 were labeled as generalized motor and 1,133,490 were labeled as non-motor generalized/focal seizures.

were recorded by 12,402 users with a recognizable seizure description (Fig. 1). Demographic information is provided in Table 1. This platform is open to general population and thus, users were not pre-screened by a neurologist.

2.2. Data pre-processing

Seizure descriptions recorded by SeizureTracker.com are shown in Table 2. Seizures descriptions were grouped into generalized motor seizures (“tonic-clonic” or “secondarily generalized”), generalized non-motor/focal seizures, or non-coded seizures (“other”, “unknown”, or missing). Inclusion criteria included seizure entries with either generalized motor or generalized non-motor/focal seizure types. Seizures with missing descriptions or recorded as “other” or “unknown” type were excluded. The number of generalized motor seizures was calculated per user for each year since the initial seizure diary entry. Based on prior literature, each year was then categorized as one of three risk stratification groups: Low (0 generalized motor seizures/year), Medium (1–2 generalized motor seizures/year), or High (≥ 3 generalized motor seizures/year) [7]. The Low risk group includes users with generalized non-motor (e.g. absence) or focal seizures, but with no generalized motor seizures. For all users, for each year with no generalized motor seizures recorded, but at least 1 seizure of any kind recorded in a subsequent year, the number of generalized motor seizures was set to zero and recorded as “Low.” Thus, in order to be included in the study, users had to have documented at least one seizure of any type, but risk stratification was based on number of GTC/year, so Medium and High risk users that did not record GTC in a year but did record other seizure types were assumed to be free of generalized motor seizures and shifted into the Low risk group.

2.3. Statistical analysis

For each user, an initial rate category was assigned based on the first diary year. Duration of analysis was limited to the duration of the group with minimal duration of recording (3 years; medium risk group). Users were considered “stable” in their rate category for as many years as the same category remained unchanged. Low, Medium, and High rate groups were analyzed with Kaplan-Meier analysis to determine the proportion remaining within initial rate category per year. A survival trend test (global log-rank test) was used to test for an overall significant difference between groups and pairwise log-rank tests were used to test for significant differences between individual groups. Statistical significance was evaluated at the $\alpha = 0.05$ level, and corrected for multiple comparisons using the Bonferroni correction. Analysis was performed in Tableau (v10.5) and R (v3.1.3).

3. Results

Of the 1.4 million seizures analyzed, 19% of seizures (42.8% of all users) were generalized motor seizures. The 12,402 users were divided

into low (7348 users), medium (2196 users), and high (2858 users) risk groups. The low-risk group had a larger number of users compared to medium- and high-risk groups. Mean age and gender distribution was similar among the three groups. In terms of congenital etiology, a higher percentage of severe childhood epilepsies, i.e. Dravet syndrome and Lennox-Gastaut syndrome was present in the high-risk group compared to low- and medium-risk groups. Additional study sample characteristics are shown in Table 1.

Kaplan-Meier survival curves for the length of time within the initial risk stratification group are provided for people at initially evaluated at high, medium and low GTC rates in Fig. 2. Of the users starting at a high GTC rate, 11.9% (339 users) changed at some point to a medium GTC rate and 10.2% (292 users) changed at some point to a low GTC rate. Of the users starting at a medium GTC rate, 6.7% (148) changed at some point to a high GTC rate and 9.6% (210) changed at some point to a low GTC rate. Of the users starting at a low GTC rate, 2.0% (146) changed at some point to a medium GTC rate and 2.0% (141) changed at some point to a high GTC rate. A significant difference in time to a departure from the initial GTC risk group was identified between the three groups ($p < 0.01$). At three years, 40.8% of people initially presenting as high-risk and 5.1% of people initially presenting as medium-risk remained in their initial stratification, while 77.3% of people initially presenting as low-risk remained in their initial stratification. Pairwise comparisons of groups revealed a significant difference between high- and low-risk groups ($p < 0.01$), medium- and low-risk groups ($p < 0.01$) and high- and medium-risk groups ($p < 0.01$).

4. Discussion

In this study, we investigated the natural history of GTC rates and provided quantitative estimates that may be used to prognosticate how rates change over time for people with self-reported seizures. We found that the vast majority of the people who initially presented with low GTC rates remained with a low rate, whereas people who initially presented with medium or high GTC rates departed from their initial level by changing groups. Higher GTC rates are significantly associated with higher SUDEP risk, high risk for injuries, and reduced quality of life [7,4,6]. If so, the present data suggests that patients’ risk factors for SUDEP should be evaluated yearly and counseled appropriately.

Usage of SeizureTracker.com data permits large-scale analysis of seizure patterns, based on one of the largest existing databases of patient-reported seizures in the world [1,2]. This database can provide in-depth understanding of natural history, as users report demographic and clinical information, such as circadian timing of seizures, age of seizure onset, and anti-seizure medication usage [9–11]. However, the self-reported nature of this database lends itself to specific limitations, including misdiagnosis, misclassification of seizure types, miscounting, or lack of patient/caregiver recognition of seizures. Studies comparing electrographic to patient-reported seizures have also found that patient-reported seizure diaries may over- or under-estimate the number of seizures [10,11]. For instance, in this study, all events, regardless of

Table 1

Baseline characteristics of study sample. Mean (SD) (for continuous variables) or percentage (for categorical variables) are shown.

	Low GTC rate (n = 7348)	Medium GTC rate (n = 2196)	High GTC rate (n = 2858)	All users (n = 12,402)
Age at initial entry diary entry, in years	20.8 (17.3)	20.8 (15.3)	17.0 (13.6)	19.9 (16.3)
Sex, female	53.2% ^a	51.5% ^b	49.0% ^c	51.9% ^d
Epilepsy etiology				
Tuberous sclerosis complex	4.4%	1.8%	2.3%	3.5%
Lennox-Gastaut syndrome	1.9%	2.7%	4.6%	2.7%
Dravet syndrome	1.1%	4.3%	7.3%	3.1%
Down syndrome	0.5%	0.2%	0.4%	0.4%
Aicardi syndrome	0.4%	0.3%	0.6%	0.4%
Rett syndrome	0.4%	0.3%	1.0%	0.5%
Angelman's syndrome	0.3%	0.2%	0.3%	0.3%
Sturge-Weber syndrome	0.3%	0.2%	0.2%	0.2%
Neurofibromatosis	0.2%	0%	0.1%	0.1%
Hypothalamic Hamartoma	0.08%	0%	0.1%	0.07%
Phelan-McDermid syndrome	0.04%	0.1%	0.04%	0.06%
CNS tumors	4.1%	2.3%	1.8%	1.8%
Traumatic brain injury	9.6%	8.4%	7.2%	7.2%
CNS infection	4.7%	3.9%	4.6%	4.6%
Stroke	3.2%	2.1%	1.9%	1.9%
Perinatal hypoxia	3.2%	3.6%	2.6%	2.6%
Maternal drug or alcohol abuse during pregnancy	0.5%	0.5%	0.4%	0.4%
Alcohol or drug abuse	0.6%	0.6%	0.3%	0.3%
High fever	2.1%	2.3%	1.7%	1.7%
CNS malformation	5.0%	3.1%	4.3%	4.3%
Other	6.6%	6.2%	7.7%	6.8%
Seizure type				
Simple partial				19.1%
Generalized tonic clonic				17.7%
Complex partial				16.7%
Tonic				16.4%
Unknown				5.5%
Absence				4.4%
Myoclonic				4.7%
Other				2.9%
Atonic				2.7%
Secondary generalized				2.4%
Aura only				2.1%
Myoclonic cluster				1.9%
Infantile spasms				1.8%
Atypical absence				0.8%
Clonic				0.6%
Gelastic				0.1%
Seizure frequency, yearly ^e				
Generalized motor seizures	0.01 (0.4)	2.0 (2.9)	306.7 (1677)	148.7 (1004.1)
All seizures	107 (663.7)	82.1 (718.8)	90.2 (388.3)	21.2 (190.2)
Circadian timing of generalized motor seizures ^e				
Nocturnal (7p-7a)	52.0%	45.6%	46.7%	46.7%
Waking (7a-7p)	48.0%	54.4%	53.3%	53.3%

^a 42.8% male; 4.0% not reported.

^b 44.7% male; 3.8% not reported.

^c 48.9% male; 2.1% not reported.

^d 44.5% male; 3.5% not reported.

^e Over first year of recording.

reported seizure duration, were included, and may include non-epileptic events. Although this decision may introduce bias in the results, this decision was made because inaccurate recording of the specific characteristics of a seizure does not invalidate the presence or absence

of a seizure event. Patients may often correctly document the occurrence of seizures in their electronic diaries without entering valid or complete information about the characteristics of the seizure itself, in order to have appropriate daily/weekly/monthly counts to give to clinicians [10,11]. Lastly, the analysis of electronic diary data but lead to sampling bias, as usage of electronic diaries requires consistent access to the internet as well as a smartphone or computer, and may thereby exclude patients of lower socio-economic status, who are at higher risk for SUDEP [12].

Several assumptions were made for the analysis. Years where no generalized motor seizures were recorded were labeled as low-risk, which may have resulted in earlier category switching events for people initially assigned to high- and medium-risk groups. Analysis of history beyond three years past the initial diary entry was limited by the small sample size of the medium-risk group ($n = 4$). Furthermore, since the years were determined based off of each user's initial date of entry and due to the timing of data collection, the final year documented may include less than twelve months, potentially resulting in early censorship for medium groups.

In translating the results of this work to prognostication for SUDEP risk, there are several additional factors which ought to be considered, including nocturnal seizures [7]. Lack of GPS location and polysomnographic or concurrent EEG recording makes analysis of nocturnal seizures challenging. Future work would benefit from the use of wearable biosensors that increases certainty about presence and absence of events at any time of day or night, as well as information about sleep time [13]. In addition to other risk factors, confounders were not accounted for in the survival analysis, so that differences in survival between risk groups may be partially attributable to baseline differences in demographic or seizure characteristics. For instance, there is a lower percentage of severe childhood epilepsies in the low-risk group, compared to the high-risk group. Anti-seizure medications are less reliably recorded in the SeizureTracker.com database, and therefore subgroup analysis to evaluate the association of anti-seizure medication use with natural history transitions is not explored here. However, future work may explore demographic differences, seizure characteristics, anti-seizure drugs, as well as how presence of nocturnal seizures should be best combined with GTC frequency in order to produce a more fine-grained risk stratification.

Despite these limitations, it is helpful to consider that perhaps a patient at low-risk for SUDEP is likely to remain low-risk (based on GTC rates), whereas higher risk patients (defined as at least one GTC/year) should be re-assessed for risk categorization yearly, as they are expected to change risk category over time. Additional research may be useful for examining whether similar patterns of SUDEP risk natural history are present among physician-verified seizures and among electrographic seizures.

Disclosures

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Declaration of Competing Interest

The other authors have no conflicts of interest to disclose.

We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

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Table 2

Classification of SeizureTracker.com descriptions into seizure types. Users have the option to describe each documented seizure, and can choose one of the following descriptions: “Absence”, “Atonic”, “Atypical Absence”, “Aura only”, “Clonic”, “Complex partial”, “gelastic”, “infantile spasm”, “myoclonic”, “myoclonic cluster”, “simple partial”, “tonic”, “tonic-clonic”, “secondarily generalized”, “other”, “unknown”; or they can choose not to describe the event (i.e. null). These descriptions were grouped as one of the following types for the purposes of this study: generalized motor, non-motor generalized/ focal, or non-coded events.

Seizure type	Seizure description (SeizureTracker.com)
Generalized motor	“Tonic-Clonic”, “Secondarily generalized”
Non-motor generalized/focal	“Absence”, “Atonic”, “Atypical Absence”, “Aura only”, “Clonic”, “Complex partial”, “gelastic”, “infantile spasm”, “myoclonic”, “myoclonic cluster”, “simple partial”, “tonic”
Non-coded events	“Other”, “Unknown”, null (not coded)

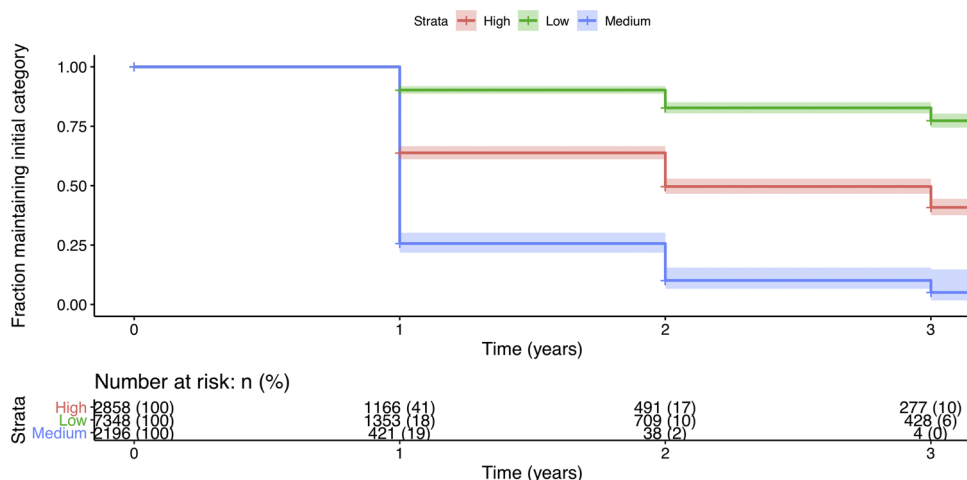


Fig. 2. SUDEP Risk Categories over Time. Kaplan-Meier survival curves for the length of time within initial GTC rate stratification group are shown for people with reported seizures at High (red), Medium (blue), and Low (green) GTC rates. The 95% confidence intervals are shaded in for each category. The Low group maintained the highest fraction, followed by High, with the Medium group showing the lowest fraction. For High, Medium, and Low stratification groups, the number of people per group (N) and the percentage remaining at time (years) after initial risk stratification A statistically significant difference was identified between treatment groups, with pairwise analysis indicating differences between high/low risk groups, high/medium and between medium/low groups. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

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References

- Casassa C, Rathbun Levit E, Goldenholz DM. Opinion and special articles: self-management in epilepsy: web-based seizure tracking applications. *Neurology* 2018;91(21):2027–30. <https://doi.org/10.1212/WNL.0000000000006547>.
- Ferastraorau AV, et al. Characteristics of large patient-reported outcomes: where can one million seizures get us? *Epilepsia Open* 2018;0–2. <https://doi.org/10.1002/epi4.12237>.
- Harden CL, et al. The effect of seizure severity on quality of life in epilepsy. *Epilepsy Behav* 2007;11:208–11. <https://doi.org/10.1016/j.yebeh.2007.05.002>.
- de Boer HM, Mula M, Sander JW. The global burden and stigma of epilepsy. *Epilepsy Behav* 2008;12(4):540–6. <https://doi.org/10.1016/j.yebeh.2007.12.019>.
- Yoon D, et al. Economic impact of epilepsy in the United States. *Epilepsia* 2009;50:2186–91. <https://doi.org/10.1111/j.1528-1167.2009.02159.x>.
- de la Loge C, et al. PatientsLikeMe® online epilepsy community: patient characteristics and predictors of poor health-related quality of life. *Epilepsy Behav* 2016;63:20–8. <https://doi.org/10.1016/j.yebeh.2016.07.035>.
- Harden C, Tomson T, Gloss D, Buchhalter J, Cross JH, Donner E, et al. Practice guideline summary: sudden unexpected death in epilepsy incidence rates and risk factors: report of the Guideline Development, Dissemination, and Implementation Subcommittee of the American Academy of Neurology and the American Epilepsy Society. *Neurology* 2017;88(17):1674–80. <https://doi.org/10.1212/WNL.0000000000003685>.
- DeGiorgio CM, et al. Ranking the leading risk factors for sudden unexpected death in epilepsy. *Front Neurol* 2017. <https://doi.org/10.3389/fneur.2017.00473>.
- Goldenholz DM, et al. Different as night and day: patterns of isolated seizures, clusters, and status epilepticus. *Epilepsia* 2018;59(5):73–7. <https://doi.org/10.1111/epi.14076>.
- Karoly P, Goldenholz DM, Cook M. Are the days of counting seizures numbered? *Curr Opin Neurol* 2018. <https://doi.org/10.1097/WCO.0000000000000533>.
- Karoly PJ, et al. Circadian and circaseptan rhythms in human epilepsy: a retrospective cohort study. *Lancet Neurol* 2018;17(11):977–85. [https://doi.org/10.1016/S1474-4422\(18\)30274-6](https://doi.org/10.1016/S1474-4422(18)30274-6).
- Kaiboriboon K, et al. Premature mortality in poor health and low income adults with epilepsy. *Epilepsia* 2014;55(11):1781–8. <https://doi.org/10.1111/epi.12789>.
- Onorati F, et al. Improvement of a convulsive seizure detector relying on accelerometer and electrodermal activity collected continuously by a wristband. *Epilepsy pipeline conference*. 2016.